Yael Vinker

+972(0)-509800322 | yael.vinker@weizmann.ac.il | yael-vinker.github.io/website

Research Areas: Computer Vision & Graphics • Machine Learning • Generative Models (AI) • Visual Communication

Academic Positions

Senior Scientist (Assistant Professor), Weizmann Institute of Science 2026–Present
Department of Computer Science and Applied Mathematics

Postdoctoral Associate, MIT CSAIL 2024–2026
Host: Prof. Antonio Torralba

Education

Ph.D. in Computer Science 2021–2024
Tel Aviv University, Israel
Advisors: Prof. Daniel Cohen-Or and Prof. Ariel Shamir
Dissertation: "Generative Visual Communication in the Era of Vision-Language Models"

M.Sc. in Computer Science 2019–2021
The Hebrew University of Jerusalem, Israel
Advisor: Prof. Raanan Fattal
Thesis: "Unpaired Learning for High Dynamic Range Image Tone Mapping"

B.Sc. in Computer Science and Visual Communication 2016–2019
The Hebrew University of Jerusalem and Bezalel Academy of Arts and Design, Israel
Magna cum laude. GPA: 94/100

Awards & Honors

Excellence and Recognition Awards

Rising Star in EECS, MIT, 2024 (link)

Blavatnik Prize for Outstanding PhD Students in Computer Science, 2024
Awarded to 4 Ph.D. students annually

Outstanding Reviewer, CVPR 2025

Paper Awards

Honorable Mention Award, ACM Transactions on Graphics (TOG), SIGGRAPH 2026
Inspiration Seeds: Learning Non-Literal Visual Combinations for Generative Exploration

Best Paper Award, ACM Transactions on Graphics (TOG), SIGGRAPH Asia 2023
Concept Decomposition for Visual Exploration and Inspiration

Honorable Mention Award, ACM Transactions on Graphics (TOG), SIGGRAPH 2023
Word-as-Image for Semantic Typography

Best Paper Award, ACM Transactions on Graphics (TOG), SIGGRAPH 2022
CLIPasso: Semantically-Aware Object Sketching

Fellowships & Scholarships

Eric and Wendy Schmidt Postdoctoral Award, 2024

Tel Aviv University President's Award for Outstanding Postdoctoral Students Abroad, 2024

Nominated for the Fulbright Postdoctoral Fellowship, 2024

VATAT Scholarship for Outstanding Female PhD Candidates, 2023
Awarded to 3 PhD students annually

Teaching

Lecturer, *Tel Aviv University* Spring 2023, Spring 2024
Deep Neural Networks for Computer Graphics (0368308901)
Main lecturer. Developed the course syllabus and presentations, and managed logistics
The course had 50-60 participants each year and received high ratings from students (6.89/7)

Teaching Assistant, *Tel Aviv University* 2022–2024
Seminar in Neural Networks-Based Graphics (0368371301)

Lecturer, *Reichman University* July 2024
Workshop on Generative AI and Diffusion Models
Main lecturer for 20 graduate students. Developed the course syllabus and presentations, and managed logistics

Work & Research Experience

Google Research 2022–2023
Student Researcher, Creative Camera team, Google Tel Aviv
Worked under Dr. Yael Pritch Knaan. Researcher in the field of generative models and VLMs

École Polytechnique Fédérale de Lausanne (EPFL) Summer 2021
Research Intern, Summer@EPFL, Switzerland
Supervised by Prof. Amir Zamir at the Visual Intelligence and Learning Lab (VILAB)

Disney Research Spring 2021
Research Intern, DisneyResearch|Studios, Zurich
Supervisors: Dr. Christopher Schroers and Dr. Roberto Azevedo
Developed content-aware video compression models

Software Engineering Intern, Google
LiveResults Proactive team, Google Tel Aviv July 2019–March 2020
OneGoogle team, Google Tel Aviv August 2018–January 2019

Grants & Funding

Designing for Sustainability MIT-HPI Research Program 2025–2026
Awarded $200,000
Project: Developing Whiteboard-Literate AI for Equitable and Accessible Education. Contributed to the project concept and proposal development. Awarded under PIs Prof. Antonio Torralba and Prof. Gerard de Melo

BioRender Research Funding 2025–2026
Support for Visual Communication Research.
Industry collaboration on generating scientific illustrations.

Invited Talks & Seminars

What Makes a Visual Idea? Decomposing and Recombining Visual Concepts
Keynote at the Human-AI Co-Creation in Generative Art Workshop SIGGRAPH 2026
Keynote at the 3rd Workshop on Visual Concepts CVPR 2026

AI for Visual Arts
Keynote at the second AI for Visual Arts Workshop ICCV 2025

Bridging Generative Models and Visual Communication
Brown University, BVC (Brown Visual Computing) December 2025
University of Toronto, Graphics Seminar December 2025
TTIC Young Researcher Seminar November 2025

Text-to-Image Personalization for Visual Design
AI Safety & Quality Seminar at Adobe — December 2025

Computational Models of Line Drawings: From Perception to Synthesis
TTIC Graduate Course on Generative Models, Art, and Perception — October 2025

AI for Visual Arts
Keynote at the second AI for Visual Arts Workshop — ICCV 2025

Personalization for Creative Applications
Keynote at P13N: Personalization in Generative AI Workshop — ICCV 2025

Visual Abstraction, Symbolism, and Narrative: Lessons from Comics for AI
Keynote at COMIQ: Comic Intelligence Quotient — ICCV 2025

Synthetic Data in Creative Applications
Keynote at SynData4CV Workshop on Synthetic Data for Computer Vision — CVPR 2025

Generative Models for Visual Communication
Runway AI — May 2025
MIT Creative Machine Learning for Design Course — April 2025

Can AI Really Sketch? Advances in Generative Models for Sketching
MIT CSAIL Vision-Graphics Seminar — April 2025

Challenges and Advancements in Generative Models for Sketches
MIT Face to Face https://artfinity.mit.edu/event/face-to-face — April 2025

Generative Models for Visual Communication
AI Vision Seminar, MIT Media Lab — November 2024
CMU BIG Seminar (roBot Intelligence Group) — November 2024

COGGRAPH Workshop (link)
An interdisciplinary workshop on the intersection of cognitive science and computer graphics — July 2024

Generative Visual Communication in the Era of Vision-Language Models
Blavatnik Prize for Outstanding Israeli Doctoral Students in Computer Science — June 2024

Generative Models for Creative Applications
MIT CSAIL Vision-Graphics Seminar — March 2024
MIT Media Lab — March 2024
Graphics Cafe Seminar, Stanford University — March 2024
Berkeley Computer Vision Seminar — March 2024
Google VisCAM Seminar — March 2024
Adobe Research GenTech Seminar — March 2024

Generative Models for Abstractions
Weizmann Vision and AI Seminar — January 2024

CLIPasso and Word-As-Image
Microsoft Data Science Bond — May 2023
Lightricks Research Seminar — March 2023

CLIPasso: Semantically-Aware Object Sketching
Israeli Computer Vision Day (link to recording) — January 2023

Interviews & Media Coverage

ACM SIGGRAPH Blog — July 2026
Thinking in Lines: How Abstraction Shapes Communication and Creativity (link to article)

Schmidt Sciences, Impact Report — 2025
Feature on my research in visual communication and sketches.
"The Architecture of Thought" (link to article)

MIT CSAIL News June 2025
"Teaching AI Models the Broad Strokes to Sketch More Like Humans Do" (link to article)

MIT CSAIL News June 2025
Featured my work in the EECS Rising Stars article (link)

Talking Papers Podcast March 2023
Interview about CLIPasso with Itzik Ben Shabat (link)

ACM SIGGRAPH Blog November 2022
Interview about CLIPasso: "Abstraction: The Key to Semantic Sketching" (link to article)

RSIP Vision Magazine October 2022
Featured on the cover page with a full interview about CLIPasso (link to article)

Academic Service

Workshop Co-Organizer, SIGGRAPH 2026
Lines and Minds: Visual Abstraction in Art, Psychology, and Computer Graphics
(https://lines-and-minds.github.io/).

Technical Papers Committee. SIGGRAPH 2025, 2026

Workshop Co-Organizer, SIGGRAPH 2025
Drawing and Sketching: Art, Psychology, and Computer Graphics (https://lines-and-minds.github.io/2025/).

Publicity Co-Chair, ICCV 2025

Reviewer for CVPR, ICCV, ECCV, SIGGRAPH, SIGGRAPH Asia, IEEE Transactions on Visualization and Computer Graphics (TVCG), ACM Transactions on Graphics (TOG)

MIT Face to Face Co-Organizer, 2025
An MIT-wide community drawing project inspired by Es Devlin, recipient of the 2025 Eugene McDermott Award in the Arts. https://artfinity.mit.edu/event/face-to-face

Mentor at the "Meduyakot" Program, Tel Aviv University, 2022–2024
Mentored female graduate students to promote leadership and inclusion of women in exact sciences

Publications

* Denotes equal contribution.

SIGGRAPH and **SIGGRAPH Asia** are the premier venues in computer graphics. Their journal-track papers are published in **ACM Transactions on Graphics (TOG)**, the flagship journal in the field (h5-index 114, ranked #1). **CVPR**, **ICCV**, and **ECCV** (h5-indices of 450, 256, and 262 respectively) are the top three venues in computer vision, with CVPR ranked #1 in computer science. **NeurIPS** (h5-index 371) is ranked #1 in artificial intelligence.

[1] Moriya Maya, Raab Sigal, **Vinker Yael**, and Dekel Tali. "FoldingAgent: Inferring Parametric Origami Procedures from Demonstration Videos". *ACM SIGGRAPH Conference Papers*. 2026.

[2] Xiaodan Du, Ruize Xu, David Yunis, **Yael Vinker**, and Greg Shakhnarovich. "Teaching an Agent to Sketch One Part at a Time". *European Conference on Computer Vision (ECCV)*. Springer. 2026, pp. 248–265.

[3] Kfir Goldberg, Elad Richardson, and **Yael Vinker**. "Inspiration Seeds: Learning Non-Literal Visual Combinations for Generative Exploration". *ACM SIGGRAPH Conference Papers*. 2026. **Best Paper Honorable Mention Award**.

[4] Mia Tang, **Yael Vinker**, Chuan Yan, Lvmin Zhang, and Maneesh Agrawala. "Instance Segmentation of Scene Sketches Using Natural Image Priors". *ACM SIGGRAPH Conference Papers*. 2025.

[5] **Yael Vinker**, Tamar Rott Shaham, Kristine Zheng, Alex Zhao, Judith E Fan, and Antonio Torralba. "SketchAgent: Language-Driven Sequential Sketch Generation". *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*. 2025, pp. 23355–23368.

[6] Ellie Arar, Yarden Frenkel, Daniel Cohen-Or, Ariel Shamir, and **Yael Vinker**. "SwiftSketch: A Diffusion Model for Image-to-Vector Sketch Generation". *ACM SIGGRAPH Conference Papers*. 2025.

[7] Sagi Polaczek, Yuval Alaluf, Elad Richardson, **Yael Vinker**, and Daniel Cohen-Or. "NeuralSVG: An Implicit Representation for Text-to-Vector Generation". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025.

[8] Rinon Gal*, **Yael Vinker***, Yuval Alaluf, Amit Bermano, Daniel Cohen-Or, Ariel Shamir, and Gal Chechik. "Breathing Life Into Sketches Using Text-to-Video Priors". *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2024, pp. 4325–4336. *Denotes equal contribution. **Spotlight Presentation (top 10%)**.

[9] Yarden Frenkel, **Yael Vinker**, Ariel Shamir, and Daniel Cohen-Or. "Implicit Style-Content Separation Using B-LoRA". *Proceedings of the 18th European Conference on Computer Vision (ECCV)*. 2024.

[10] **Yael Vinker**, Andrey Voynov, Daniel Cohen-Or, and Ariel Shamir. "Concept Decomposition for Visual Exploration and Inspiration". *ACM Transactions on Graphics (TOG)*. 2023. **Best Paper Award**.

[11] Omri Avrahami, Amir Hertz, **Yael Vinker**, Moab Arar, Shlomi Fruchter, Ohad Fried, Daniel Cohen-Or, and Dani Lischinski. "The Chosen One: Consistent Characters in Text-to-Image Diffusion Models". *ACM SIGGRAPH Conference Papers*. 2024.

[12] Kushin Mukherjee*, Holly Huey*, Xuanchen Lu*, **Yael Vinker**, Rio Aguina-Kang, Ariel Shamir, and Judith Fan. "SEVA: Leveraging sketches to evaluate alignment between human and machine visual abstraction". *Advances in Neural Information Processing Systems (NeurIPS)*. 2023. *Denotes equal contribution.

[13] Shir Iluz*, **Yael Vinker***, Amir Hertz, Daniel Berio, Daniel Cohen-Or, and Ariel Shamir. "Word-As-Image for Semantic Typography". *ACM Transactions on Graphics (TOG)*. 2023. *Denotes equal contribution. **Best Paper Honorable Mention Award**.

[14] **Yael Vinker**, Yuval Alaluf, Daniel Cohen-Or, and Ariel Shamir. "CLIPascene: Scene Sketching with Different Types and Levels of Abstraction". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2023. **Oral presentation (top 1.68%)**.

[15] Hila Chefer, Yuval Alaluf, **Yael Vinker**, Lior Wolf, and Daniel Cohen-Or. "Attend-and-Excite: Attention-Based Semantic Guidance for Text-to-Image Diffusion Models". *ACM Transactions on Graphics (TOG)*. 2023. *Denotes equal contribution.

[16] **Yael Vinker**, Ehsan Pajouheshgar, Jessica Y. Bo, Roman Christian Bachmann, Amit Haim Bermano, Daniel Cohen-Or, Amir Zamir, and Ariel Shamir. "CLIPasso: semantically-aware object sketching". *ACM Transactions on Graphics (TOG)* 41.4 (2022). ISSN: 0730-0301. **Best Paper Award**.

[17] **Yael Vinker***, Eliahu Horwitz*, Nir Zabari, and Yedid Hoshen. "Image Shape Manipulation From a Single Augmented Training Sample". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2021. *Denotes equal contribution. **Oral Presentation (top 3%)**.

[18] **Yael Vinker**, Huberman-Spiegelglas Inbar, and Fattal Raanan. "Unpaired Learning for High Dynamic Range Image Tone Mapping". *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. 2021.

Manuscripts Under Review

[1] Hui Ren, Yuval Alaluf, Omer Bar Tal, Antonio Torralba, Alexander Schwing, and **Yael Vinker**. "Versatile Sequential Sketch Generation with Video Model Priors". 2026.

[2] Marco Cipriano, Hannah Gao, Idan Shenfeld, Gerard de Melo, Antonio Torralba, and **Yael Vinker**. "Measuring and Improving Drawing Abilities in Vision-Language Models through the Lens of Human Cognition". 2026.